- The acceleration due to gravity is denoted by $g\,\mathrm{m\,s^{-2}}$. Unless otherwise instructed, when a numerical value is needed, use $g = 9.8$.

10 Rory runs a distance of 45 m in 12.5 s. He starts from rest and accelerates to a speed of $4\,\mathrm{m\,s^{-1}}$. He runs the remaining distance at $4\,\mathrm{m\,s^{-1}}$.

Rory proposes a model in which the acceleration is constant until time T seconds.

(i) Sketch the velocity-time graph for Rory's run using this model. **[2]**

(ii) Calculate T. **[2]**

(iii) Find an expression for Rory's displacement at time t s for $0 \leqslant t \leqslant T$. **[2]**

(iv) Use this model to find the time taken for Rory to run the first 4 m. **[1]**

Rory proposes a refined model in which the velocity during the acceleration phase is a quadratic function of t. The graph of Rory's quadratic goes through (0, 0) and has its maximum point at (S, 4). In this model the acceleration phase lasts until time S seconds, after which the velocity is constant.

(v) Sketch a velocity-time graph that represents Rory's run using this refined model. **[1]**

(vi) State with a reason whether S is greater than T or less than T. (You are not required to calculate the value of S.) **[1]**